

Shivanshu Tiwari

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Summary

ML Engineer / GenAI Engineer with production experience in **computer vision, object detection, deep learning**, and **LLM-powered applications**. Built and deployed **drone detection systems** (YOLOv8, custom PyTorch CNN) at **DRDO, Ministry of Defence**, achieving **0.94 mAP@0.50**. Skilled in **Python, PyTorch, FastAPI, Docker, CI/CD, LangChain, RAG**, and **MLOps**. Seeking roles in **ML Engineering, GenAI Engineering, Computer Vision**, and **AI Application Development**.

Experience

- DRDO – Institute for Research in Defence (IRDE), Ministry of Defence** May 2026 – Jul 2026
Machine Learning Intern | Computer Vision & Deep Learning Dehradun, India
- Built **SwarmTally** – full-stack AI drone detection & counting platform using **YOLOv8, FastAPI, Next.js, Docker, CI/CD**; achieved **0.94 mAP@0.50, 0.94 precision, 0.89 recall** on electro-optical surveillance dataset.
 - Engineered **DroneVision** – custom single-class object detector built **from scratch in PyTorch**: implemented **CNN backbone, Feature Pyramid Network (FPN), multi-scale detection head, CIoU loss, NMS**, data augmentation, training & evaluation pipelines without pretrained frameworks.
 - Explored defence-relevant challenges: **small-object UAV detection, multi-target counting, real-time inference**, and EO/IR sensor fusion for airspace monitoring.
- Codec Technologies India** Apr 2026 – May 2026
Artificial Intelligence Intern Remote, India
- Built **Spam Email Classifier** using **Naive Bayes + CountVectorizer**; achieved 95%+ accuracy with Scikit-learn.
 - Developed **content-based Movie Recommendation Engine** using **TF-IDF Vectorization & Cosine Similarity** (NLP).
 - Designed **Rule-Based NLP Chatbot** with intent recognition and pattern matching for automated query resolution.

Projects

- SwarmTally – AI Drone Detection & Counting System** | *YOLOv8, FastAPI, Next.js, Docker, GitHub Actions, Vercel*
- Production full-stack app: **YOLOv8 inference** via **FastAPI** backend + **Next.js/TypeScript** frontend; **Docker Compose** orchestration, **CI/CD** with GitHub Actions, deployed on **Vercel + Render**.
 - Delivers annotated images with **bounding boxes, drone count, per-detection confidence scores** in under **200ms** on GPU. [GitHub]
- DroneVision – Custom Object Detector from Scratch** | *PyTorch, FPN, CIoU Loss, MLflow, DVC, Hugging Face Spaces*
- Implemented full detection pipeline from scratch: **custom CNN backbone (~2.3M params), FPN neck, multi-scale detection head (3 scales × 3 anchors), CIoU loss**, Mosaic/MixUp augmentation – **zero pretrained weights**.
 - MLOps-ready: **MLflow** experiment tracking, **DVC** data versioning, **GitHub Actions CI, Docker**, deployed on **Hugging Face Spaces**. [GitHub]
- AlgoTrader-AI – Stock Price Prediction** | *Python, LSTM, Time-Series, Data Pipelines*
- Production-grade **LSTM-based time-series stock prediction** system with robust **data ingestion pipelines**, model training, and evaluation on real market data. [GitHub]
- MCP-Doctor – AI Diagnostic Toolkit** | *Python, MCP, AI Agents, LLM Tooling*
- Open-source CLI tool that detects, explains, and fixes broken **MCP (Model Context Protocol)** server configs, AI agent environment issues, dependencies, and transport errors. [GitHub]

Technical Skills

ML & Deep Learning: Python, PyTorch, TensorFlow, Keras, Scikit-learn, CNN, FPN, LSTM, ANN, Transfer Learning, NMS, CIoU Loss, Data Augmentation, Model Evaluation (mAP, F1, Precision, Recall)

Computer Vision: Object Detection, YOLOv8, OpenCV, Image Classification, Bounding Box Regression, Multi-Scale Detection, Small-Object Detection

GenAI & LLM: LangChain, LangGraph, OpenAI API, Hugging Face Transformers, RAG, AI Agents, Prompt Engineering, Embeddings, Vector Databases (FAISS, ChromaDB, Pinecone)

MLOps & Deployment: MLflow, DVC, Docker, Docker Compose, GitHub Actions (CI/CD), FastAPI, REST API, Vercel, Render, Hugging Face Spaces

Data Engineering: Pandas, NumPy, SQL, ETL Pipelines, Apache Airflow, Data Preprocessing, Feature Engineering

Languages & Tools: Python, TypeScript, Bash, SQL, Git, GitHub, Linux, Jupyter, Next.js (FastAPI integration)

Education

Veer Bahadur Singh Purvanchal University 2023 – Present
B.Tech – Information Technology Jaunpur, Uttar Pradesh, India

Certifications

Oracle Cloud Infrastructure 2025 – Certified Generative AI Professional	2025
Oracle Cloud Infrastructure 2025 – Certified Data Science Professional	2025
Oracle Cloud Infrastructure 2025 – Certified AI Foundations Associate	2025